Human Review: Bourgain-Root Counterexamples in ℓ_p

Original automatic proof: `solution_packet.pdf`
Checked date: 17 June 2026
Status: Manually checked.
Verdict: The proof appears correct.

Human Comments

The proof appears to correctly give a negative answer to the ℓ_p bullet of Question 5.2 in Kania–Swaczyna's paper, for every $1 < p < \infty$, assuming Bourgain's original ℓ_1 example is being used exactly as stated.

The main idea is not to use a linear embedding or a linear isomorphism between ℓ_1 and ℓ_p. Rather, the proof uses the coordinatewise nonlinear maps

$$\Phi_p(t) = \operatorname{sgn}(t)|t|^{1/p}, \qquad \Psi_p(t) = \operatorname{sgn}(t)|t|^p.$$

Applied coordinatewise, these maps give an order-preserving bijective correspondence between the underlying coordinatewise ordered sets of ℓ_1 and ℓ_p. This correspondence is emphatically not additive, but additivity is not what is needed here. The relevant structure is coordinatewise order, domination, and order convergence.

Starting with Bourgain's ℓ_1 sequence (z_n), define

$$x_n = \Phi_p(z_n).$$

Then $x_n \in \ell_p$, since

$$\|x_n\|_p^p = \sum_j |x_n(j)|^p = \sum_j |z_n(j)| = \|z_n\|_1.$$

The key technical point is that the root map transports order-null difference sequences from ℓ_1 to ℓ_p. The scalar estimate

$$|\Phi_p(s) - \Phi_p(t)|^p \le 2^{p-1}|s - t|$$

is enough for this. If $a_m - b_m \to 0$ in order in ℓ_1, then the differences are eventually dominated by positive ℓ_1-vectors $w_\alpha \downarrow 0$. The estimate gives domination in ℓ_p by $2^{1-1/p} w_\alpha^{1/p}$, and these vectors decrease to 0 in ℓ_p. Hence

$$\Phi_p(a_m) - \Phi_p(b_m) \to 0$$

in order in ℓ_p.

Therefore Bourgain's subsequence-difference Cauchy condition is transported to ℓ_p: for every increasing subsequence (n_k),

$$z_{n_{k+1}} - z_{n_k} \to 0 \text{ in order in } \ell_1$$

implies

$$x_{n_{k+1}} - x_{n_k} \to 0 \text{ in order in } \ell_p.$$

Thus (x_n) is Buo-Cauchy in ℓ_p.

The non-convergence argument also appears correct. If x_n were order convergent in ℓ_p, then it would be coordinatewise convergent and eventually dominated by some $y \in (\ell_p)_+$. Applying the inverse coordinatewise power map gives

$$z_n = \Psi_p(x_n),$$

with coordinatewise convergence and domination by $y^p \in \ell_1$, up to finitely many initial terms. Hence (z_n) would order converge in ℓ_1, contradicting Bourgain's example.

Review Outcome

Archive this packet as an apparently correct counterexample, pending any later literature or author feedback. Conceptually, the proof transfers Bourgain's ℓ_1 obstruction to every ℓ_p, $1 < p < \infty$, through the nonlinear coordinatewise order isomorphism $t \mapsto \operatorname{sgn}(t)|t|^{1/p}$. The map is not additive, but the argument only needs preservation of order-null dominated differences and the inverse domination argument.